Stéphane KPOVIESSI

Big Data & AI Engineer

Cergy, Île-de-France, France | +33 7 48 59 47 13
oastephanekpoviessi@gmail.com | LinkedIn | GitHub | Seshat-AI | Portfolio

Profile

Big Data and Data Science engineering graduate specialising in applied AI, intelligent automation and production-grade data pipelines. Experienced in critical enterprise environments through work at Allianz involving Python/PySpark migration, data quality and automated testing. Interested in building practical solutions using AI agents, RAG, NLP, chatbots and business workflow automation.

Professional Experience

Data Engineer / Data Scientist Nov 2025 – Apr 2026
Allianz France — P4D Division, Advanced Data & Climate Team *Paris, France*

- Migrated an insurance datamart from SAS to **Python/PySpark**, ensuring functional parity across more than 15,000 lines of code.
- Implemented a **Medallion architecture** on Azure Databricks, incorporating over 150 business rules to provide reliable data for analytics use cases.
- Designed and automated a **geospatial data quality** pipeline, improving the reliability of reporting and downstream processing.
- Built a SAS/Python **automated testing framework** to support regression testing, production readiness and reliable deployment.

Data Analyst – Strategic Support Missions 2023 – 2025
LYNE ET FRERES SARL (Recurring Data Analysis Missions) *Ikpinle, Benin / Remote*

- Analysed commercial, operational and market data with **Python/Pandas** to support diversification decisions and monitor key performance indicators.
- Designed reports and interactive dashboards with **Streamlit** and HTML/CSS/JavaScript interfaces to share analyses with company staff.
- Performed analytical monitoring of agricultural activities, including palm plantations and transformation stages, to identify operational improvement opportunities.
- Set up and supported responsible internal use of AI tools based on **OpenWebUI** and open-source models for business use cases.

Machine Learning Engineer May 2024 – Aug 2024
Groupe Sylvagreg (Internship) *Lille, France*

- Researched and developed a 3D reconstruction solution from 2D images using computer vision, photogrammetry and PyTorch-based generative models.
- Automated preprocessing, data augmentation and generation pipelines to improve inference reliability.
- Deployed the solution on **AWS**, using **Docker** containers and a visualisation interface for business users.

Selected Projects

Seshat & Seshat-AI | *Go, Python, PostgreSQL, RAG, MCP, AI Agents* GitHub

- Developed an automation and assistance platform based on **AI agents**, document retrieval, knowledge management and AI-driven workflows.
- Designed a modular runtime with agent orchestration, persistent memory, hybrid RAG, multi-provider LLM integration and **MCP** support for external tools.
- Implemented enterprise-oriented features including permissions, human-in-the-loop oversight, sessions, audit trails, real-time communication and scheduled tasks.

Tech Watch Agent | *Python, FastAPI, LangGraph, PostgreSQL, pgvector, React, Docker* GitHub

- Developed a **multi-agent technology intelligence platform** automating planning, multi-source collection, structured synthesis and targeted report delivery.
- Implemented a FastAPI backend with LangGraph orchestration, web/academic/code search, session persistence and semantic memory using PostgreSQL and **pgvector**.
- Productionised the Docker stack with a React frontend, runtime LLM/embedding provider configuration, recurring profiles and reusable email groups.

Solar Forecasting | *Python, Pandas, Scikit-learn, XGBoost* GitHub

- Developed **solar power forecasting models** using weather and calendar data to support energy planning.
- Compared and optimised tabular forecasting models using feature engineering and time-based variables to improve predictive quality and robustness.
- Evaluated regression metrics and feature influence to make results interpretable and actionable.

Brain Tumour Classification | *TensorFlow, Keras, CNN, ViT, Grad-CAM* Kaggle

- Developed a **deep learning** pipeline for multi-class brain MRI classification, covering image preparation, model training and validation.
- Compared **CNNs** and **Vision Transformers**, and used **Grad-CAM** to visualise the image regions influencing predictions.

Technical Skills

Programming: Python, SQL, Go, R, Bash, C
Automation & AI: LLMs, RAG, NLP, chatbots, multi-agent systems, LangGraph, LangChain, workflow automation, human-in-the-loop
Data Engineering & Analytics: PySpark, Delta Lake, ETL/ELT, dbt, Apache Airflow, Azure Databricks, Medallion architecture, data quality, Power BI
Machine Learning & Deep Learning: Pandas, Scikit-learn, XGBoost, PyTorch, TensorFlow, Keras, CNNs, Vision Transformers, forecasting, model evaluation
Backend & APIs: FastAPI, REST APIs, SSE, PostgreSQL, pgvector, external tool integration
Tools & Cloud: Git, Docker, CI/CD, monitoring, Azure (ADLS, ADF, Databricks), AWS (EC2, S3), Streamlit, React

Education

JUNIA ISEN Lille, France
French Engineering Degree (Master's level) — Big Data & Data Science *Graduated 2026*

JUNIA ISEN Rabat, Morocco
Bachelor's Degree — Electronics and Digital Systems *2023*

Marie Stella Preparatory Programme Cotonou, Benin
MPSI Preparatory Classes — Mathematics, Physics and Engineering Science *2022*

Languages

French: Native | **English**: Professional proficiency (B2, Linguaskill Business English certified)
Yoruba, Fon, Goun: Native / bilingual

Leadership & Volunteering

Vice-President Sep 2024 – Aug 2025
JUNIA ALUMNI Network Committee *Lille, France*

- Coordinated the committee, engaged the alumni network and helped organise after-work events and networking sessions.
- Represented the association to students and coordinated joint initiatives with the Students' Union.

Communications Lead Sep 2023 – Aug 2024
JUNIA ALUMNI Network Committee *Lille, France*

- Led association communications, including announcements, event promotion and community updates.
- Created communication materials and increased the visibility of initiatives to strengthen alumni engagement.